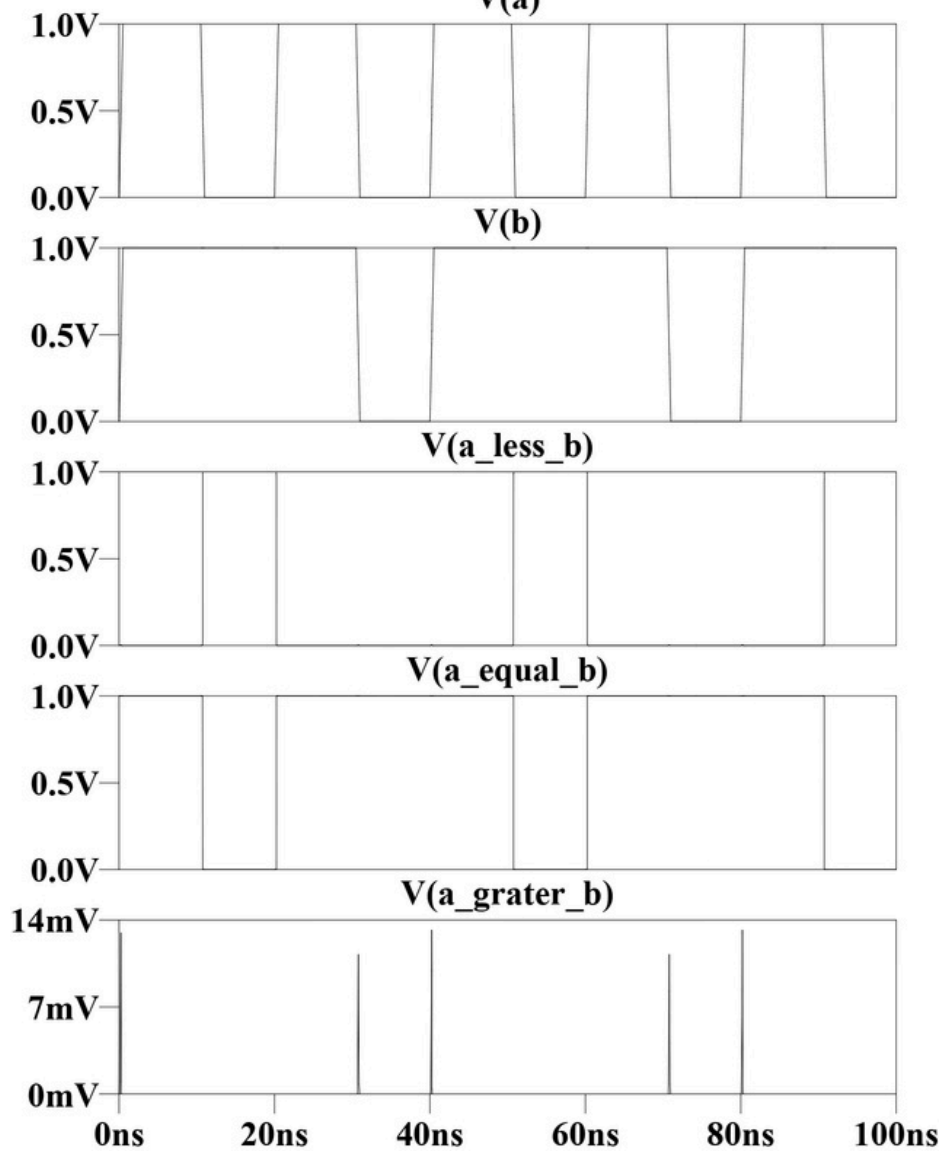
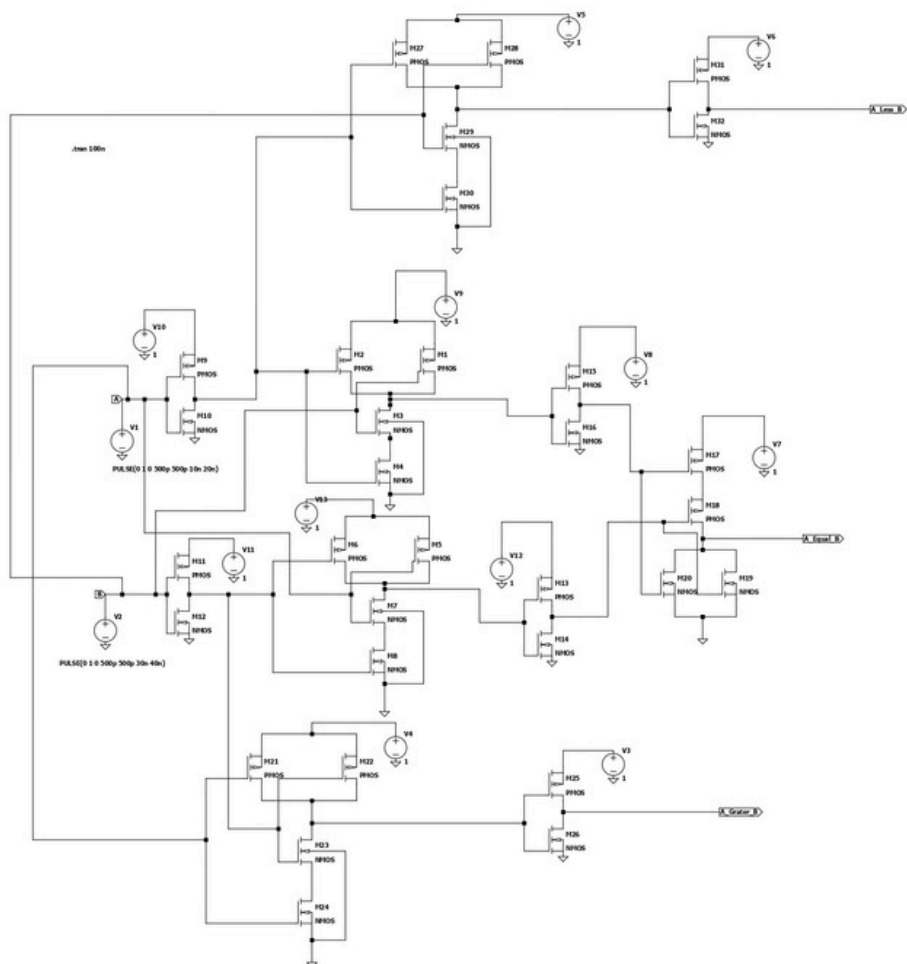


Static

Operating point DC

v(A_less_B) V = 96.139489nV

V(A_Equal_B) V = 1.799999V

V(A_Grater_B) V = 96.139502nV

Voltage = 1.8v

Transistor Count = 32

Rise Time Fall Time & Differentiate

(A_Less_B) rise time 50.79ns voltage = 544.64mV

Fall Time 609.27297 voltage = 211.81676mV

Differ 9.3264ns voltage = - 718.36mV

Frequency = 107.2222MHZ

(A_Equal_B) rise time 50.79ns voltage = 1.326124V

Fall Time 60.0456 ns voltage = 134.094mV

Differ 30.6994382ns voltage = -1.475435V

Frequency = 32.5784MHz

(A_Grater_B) risetime 30.650685 voltage = 14.71344mV

Fall Time 40.435388 ns voltage = 17.347mV

Differ 10.233167ns , volatge = 0V

fequency = 97.721579Mhz

Dc Opearting point curent & dissipation

V1 = 0A, 0W

V2 = 1A, 0W

V3 = -5.11PA, -9.20PW

V4 = -10.47PA, -18.85PW

V5 = -6.98PA, -12.57PW

V6 = -5.11PA, -9.20 PW

V7 = -10.34PA, -18.61PW

V8 = -5.11PA, -9.20PW

V9 = -6.98 PA, -12.57PW

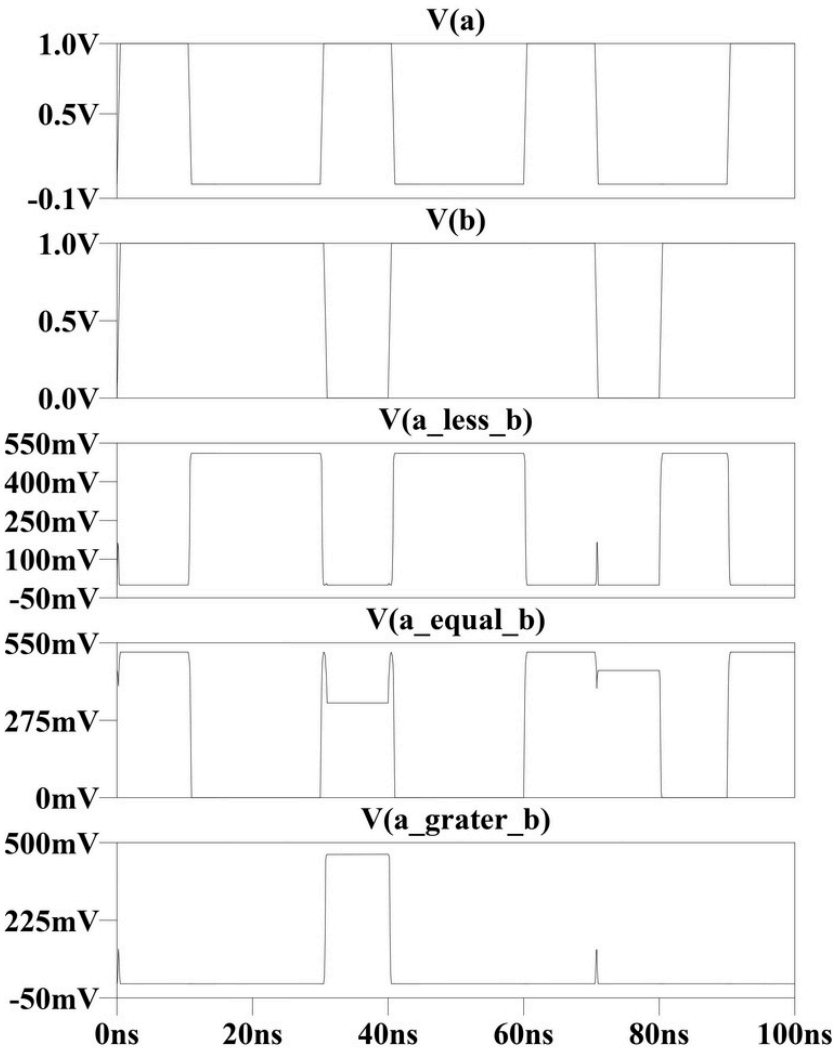
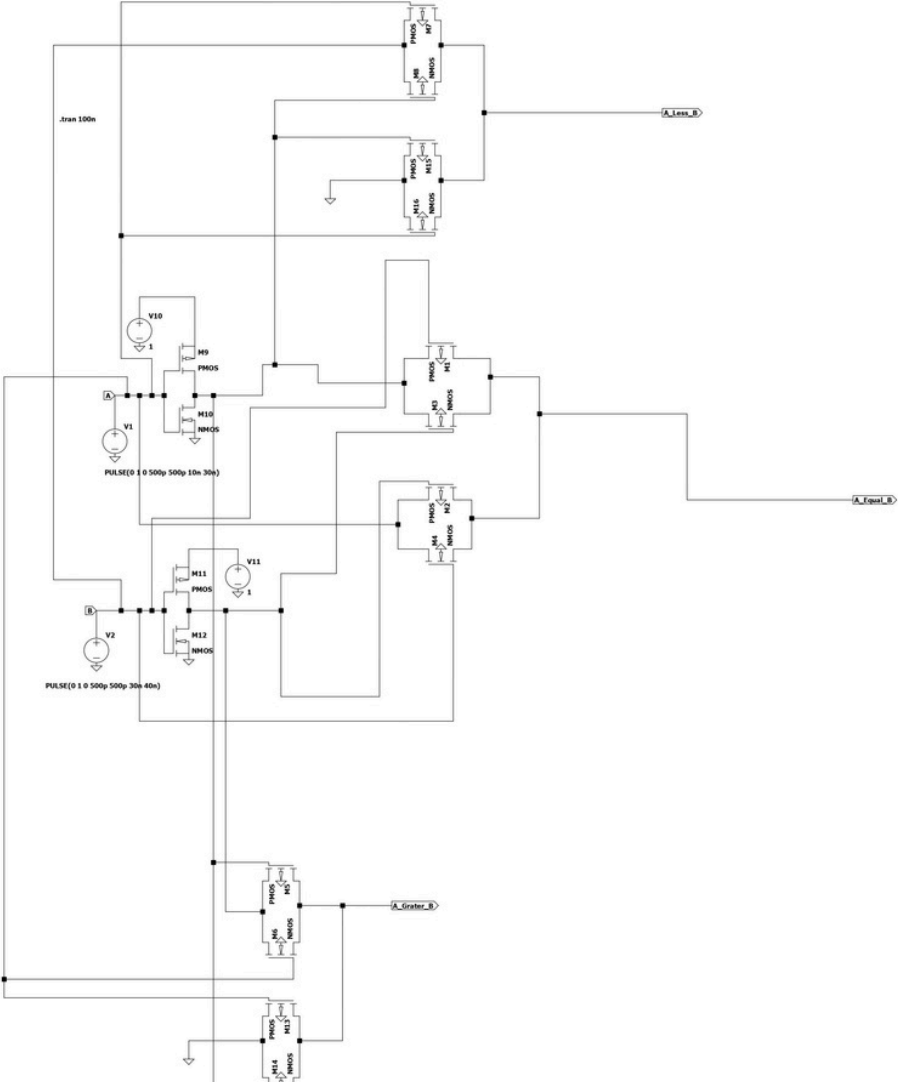
V10 = -5.17PA, -9.30PW

V11 = -5.17PA, 9.30PW

V12 = -5.11PA, 9.20PW

V13 = -6.98PA, 12.57PW

TG



TG

Operating point DC

v(A_less_B) V = 0V

V(A_Equal_B) V= 461.850667mv

V(A_Grater_B_) V = 1.071838uV

Voltage = 1.8v

Transistor Count = 16

Rise Time Fall Time & Differentiate

(A_Less_B) rise time 10.79ns voltage = 499.25mv

Fall Time 20.3115ns voltage = 260.173mV

Differ 9.29ns voltage = - 26.08mV

Frequency = 107.64 MHz

(A_Equal_B) rise time 115.40ps voltage= 427.35 mV

Fall Time 10.96 ns voltage =129.74 mv

Differ 31.16 ns voltage= 486.23mV

Frequency = 32.09MHz

(A_Grater_B) risetime 30.75ns voltage = 222.85 mV

Fall Time 40.45 ns voltage = 21.39 mV

Differ 9.52 ns , volatge =95.74mV

fequency =105.03Mhz

Dc Opearting point & Dissipation

V1 = 923.70133fA ,0W

V2 = -0A, -0W

v10 =V3 = -29.23uA, -52.62uW

V11=V4 = -29.22uA, -52.59uW

The figure displays five digital timing diagrams stacked vertically, sharing a common x-axis representing time from 0ns to 100ns. The y-axis for all plots represents voltage from 0.0V to 1.8V.

- V(a):** A square wave that is high (1.8V) from 0ns to 20ns, low (0.0V) from 20ns to 40ns, high from 40ns to 60ns, and low from 60ns to 100ns.
- V(b):** A square wave that is high (1.8V) from 0ns to 40ns, low (0.0V) from 40ns to 60ns, high from 60ns to 80ns, and low from 80ns to 100ns.
- V(a_less_b):** A signal that is high (1.8V) when V(a) is low and V(b) is high (from 20ns to 40ns), and low (0.0V) otherwise.
- V(a_equal_b):** A signal that is high (1.8V) when V(a) and V(b) are both high (from 40ns to 60ns), and low (0.0V) otherwise.
- V(a_grater_b):** A signal that is high (1.8V) when V(a) is high and V(b) is low (from 60ns to 80ns), and low (0.0V) otherwise.

PTL

Operating point DC

v(A_less_B) V = 49.999986nV

V(A_Equal_B) V = 1.79926238V

V(A_Grater_B) V = 0V

Voltage = 1.8v

Transistor Count = 10

Rise Time Fall Time & Differentiate

(A_Less_B) rise time 10.77ns voltage = 1.32V

Fall Time 20.248ns voltage = 875.77mV

Differ 9.36ns voltage = - 1.519V

Frequency = 106.80 MHz

(A_Equal_B) rise time 20.24ns voltage = 883.40 mV

Fall Time 30.19 ns voltage = 196.13 nV

Differ 9.87 ns voltage = -950.83 mV

Frequency = 101.27 MHz

(A_Grater_B) risetime 30.13ns voltage = 1.797 V

Fall Time 30.711 ns voltage = 1.037 V

Differ 1.008 ns , voltage = 1.729 V

frequency = 991.25 Mhz

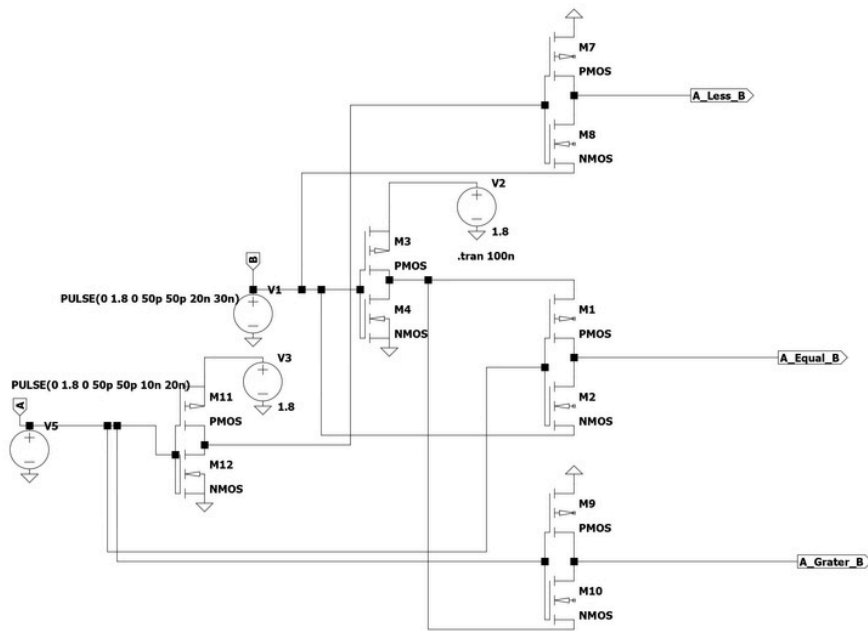
Dc Operating point & Dissipation

V1 = -5.17012PA , -9.306229PW

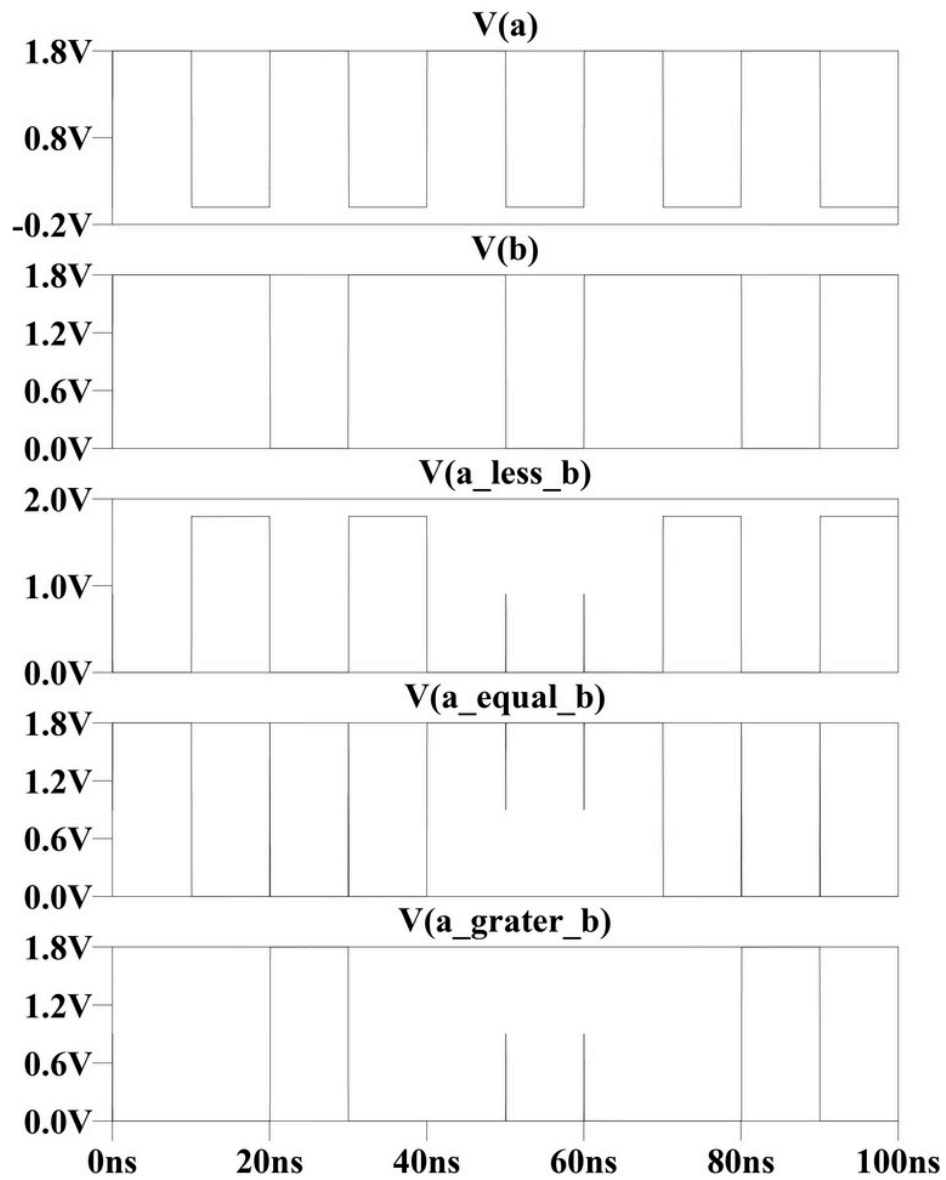
V2 = -1.7990PA, -0W

V3 = 0A, 0W

V4 = -16.007PA, -28.814244PW



GDI



GDI

Operating point DC

v(A_less_B) V = 2.997377e-27fV

V(A_Equal_B) V = 1.799995V

V(A_Grater_B) V = 578.8194uV

Voltage = 1.8v

Transistor Count = 10

Rise Time Fall Time & Differentiate

(A_Less_B) rise time 10.732ns voltage = 730.138mv

Fall Time 9.1748ns voltage = 229.49mV

Differ 9.1748ns voltage = 4.95mV

Frequency = 108.993 MHz

(A_Equal_B) rise time 20.54ns voltage= 1.79 V

Fall Time 50.843ns voltage = 5787.86 mv

Differ 29.83 ns voltage= 55.830319 mV

Frequency = 33.52MHz

(A_Grater_B) risetime 30.75ns voltage = 878.159 mV

Fall Time 30.87 ns voltage = 4863.43 mV

Differ 288.517 ps , volatge =426.71mV

fequency =3.460 GHz

Dc Opearting point & Dissipition

V1 = 3.3531035PA ,0W

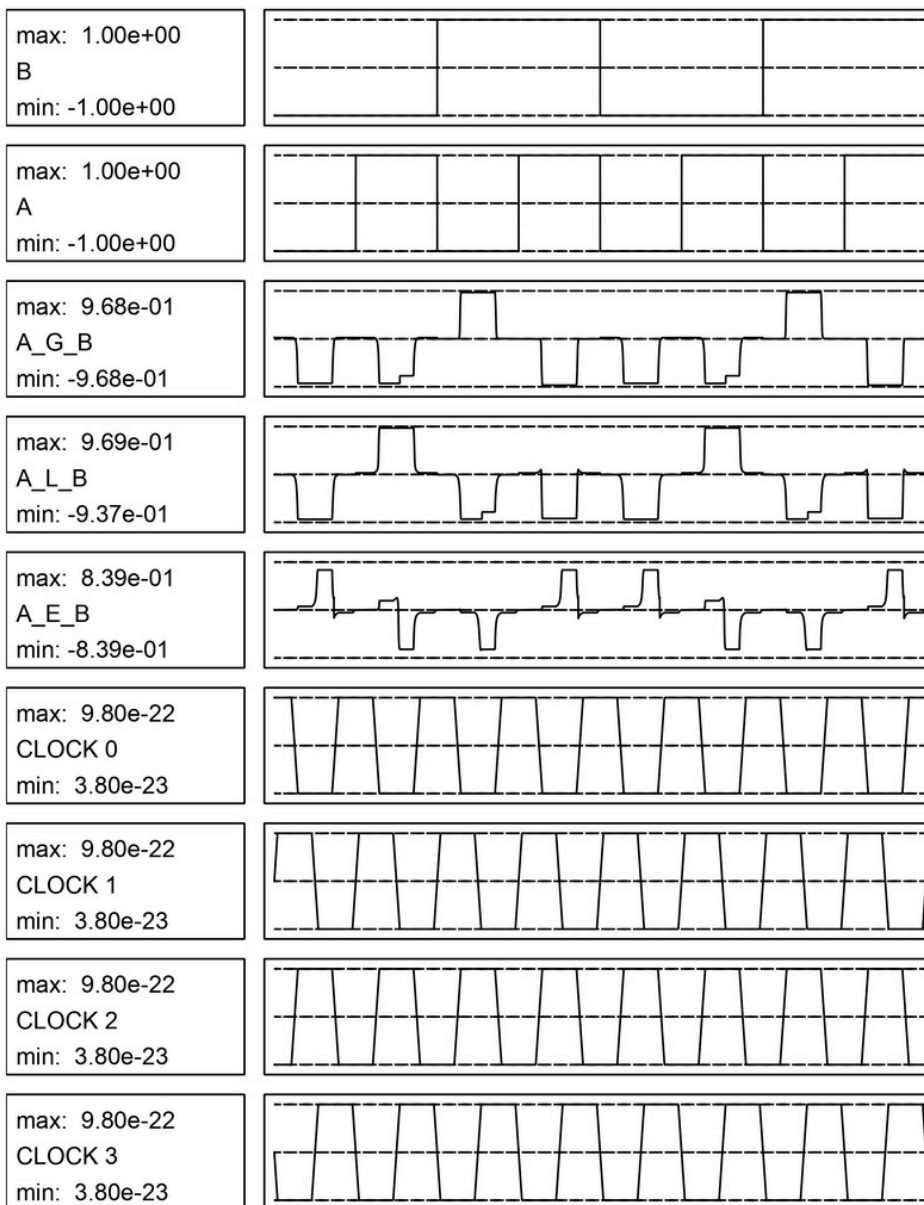
V2 = 13.820667PA, -24.8722OW

V3 = -517.01727PA,-9.30632289PW

V4 = 0A, 0W



Simulation Results



0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000 11000 12000

QCA

A=B

45nm

A = B	Static	TG	PTL	GDI
Rise Time	20.441595ns	20.260492ns	20.441595ns	20.438472ns
Fall Time	50.641026ns	30.969609ns	30.05698ns	30.268741ns
Popagation Delay	35.5413105ns	25.6159505ns	25.24922875ns	25.3536065ns

A=B

90nm

A = B	Static	TG	PTL	GDI
Rise Time	20.38835ns	20.973154ns	20.37037ns	20.438472ns
Fall Time	50ns	50.838926ns	30.05698ns	30.551627ns
Popagation Delay	35.194175ns	35.90604ns	25.213675ns	25.4950495ns

A=B

180nm

A = B	Static	TG	PTL	GDI
Rise Time	20.873786ns	20.115774ns	20.357143ns	20.438472ns
Fall Time	50ns	30.969609ns	30.071429ns	30.551627ns
Popagation Delay	35.5436693	25.540932ns	25.214286	25.4950495ns